

Kevin Elaba, M.S

Email | LinkedIn | Github | Portfolio

EDUCATION

Pittsburgh, PA

Carnegie Mellon University, School of Computer Science

- **Major:** Computational Biology, Machine Learning Concentration. M.S. (GPA: 3.5)
- **Relevant Coursework:** Bioimage Informatics, Machine Learning for Large Datasets, Applied Deep Learning

Worcester, MA

College of the Holy Cross

- **Major:** Computer Science, Pre-Medical Concentration. B.A (GPA: 3.6)
- **Relevant Coursework:** Data Structures and Algorithms, Computer Networking, Topics in Genomics

SKILLS AND CERTIFICATIONS

Certifications	AWS Machine Learning Specialty, AWS Cloud Practitioner, MIT Reinforcement Learning	
Machine Learning	(proficient): PyTorch, Scikit-learn, NumPy, SciPy, Pandas	(familiar): MLFlow, OpenCV
Cloud/Programming	(proficient): AWS (EC2, S3, Batch), Docker, Python	(familiar): Django, FastAPI

PROFESSIONAL EXPERIENCE

Software Engineer, Machine Learning	Foundation Medicine	Sep. 2023 - Present
• Integrated feature engineering and traditional machine learning inference jobs into NGS bioinformatics workflows, introducing novel classification methods into the product • Led a cost savings effort to refactor petabytes of genomic data with Python and AWS infrastructure, achieving ~\$150,000 saved annually • Modeled and implemented optimized resource allocations across distributed compute jobs, achieving a 47% improved memory efficiency • Developed and published a Python package to modularize an internal workflow language to create bioinformatics pipelines of arbitrary size, increasing research productivity by 80% • <u>Leveraged knowledge</u> in Python, Linux, PostgreSQL, Docker, AWS S3/EC2/Batch, and Git		
Research Engineer, Data Science Intern	Ampersand Biomedicines	Jan. - Jun. 2023
• Developed single-cell RNA sequencing pipelines to identify cell-type specific genes, leading to identifying marker genes from 17 new cell types • Evaluated three deep learning models for batch effect removal, achieving a 2-fold improvement in gene identification • Improved the runtime performance of the analysis by 50% • <u>Leveraged knowledge</u> in Python, AWS EC2, NumPy, and ScanPy		
Cloud Engineer, Intern	Foundation Medicine	Jun. - Aug. 2022
• Built an AWS CloudWatch dashboard to monitor 150+ alarms for real time data processing • Automated dashboard updates with AWS Lambda, deployed the tool via Docker and Terraform • Configured AWS resources and developed the Node.js API of Patiently, an interactive medical report • <u>Leveraged knowledge</u> in Python, AWS CloudWatch, Docker, Terraform, Node.js, and Git		
Genomics Research Assistant	University of Massachusetts Medical School	Jun. - Dec. 2020
• Modeled correlations between long noncoding RNA tissue types and their developmental timepoints • Built Bash scripts to compile genomic data from public atlases (ENCODE/Kaessman/NON-CODE) • Presented findings during weekly project updates and actively participated in literature review meetings • <u>Leveraged knowledge</u> in Python, R, and Bash		

MACHINE LEARNING PROJECTS

Personal Website: <https://kevinelaba.com/> for additional information and projects

Computer Vision Tissue Segmentation with U-Nets (PyTorch, OpenCV | [Technical Report](#))

- Built an in-house and fine-tuned a pre-trained U-Net model for image segmentation from H&E slides, achieving a 0.80 mean dice coefficient score
- Performed exploratory analysis, data cleaning and augmentation on AWS S3/EC2 infrastructure

Answering Jeopardy! Questions With Retrieval Augmented Generation (PyTorch | [Technical Report](#))

- Architected a LLM pipeline to answer Jeopardy questions with Wikipedia articles. Implemented RAG with a FAISS vector store and deployed the pipeline on AWS S3/EC2
- Fine-tuned Google's T5 model for question reformulation and utilized RoBERTa for question-answer inference

Modeling Antibody Conformation with Active Learning (Scikit-Learn)

- Led a 4-person ML team, designing active learning models to predict molecular stability, achieving top performance within 15% of occurrences in the sample space
- Designed the project pipeline and implemented featurization, visualization, and active learning methods